

1) LangChain

LangChain is basically a framework that helps you build applications using language models. Instead of just asking a model questions, it lets you connect the model to other tools, databases, or APIs. Think of it as a bridge between a language model and the real-world data or tasks you want it to handle. It makes AI more useful because you can chain different actions together and automate processes.

2) LLMs (Large Language Models)

LLMs, or Large Language Models, are AI models trained on massive amounts of text. They learn patterns in language and can generate human-like text, answer questions, summarize content, or even write code. The key thing is that they don't have reasoning or memory by default; they generate responses based on patterns they learned from data.

3) RAG (Retrieval-Augmented Generation)

RAG is a method where a language model doesn't just guess answers—it actually looks up relevant information first. It retrieves documents or data from a database and then uses that information to generate more accurate and specific answers. This makes AI smarter and less likely to give made-up information.

4) FAISS (Facebook AI for Semantic Search)

FAISS is a tool developed by Facebook to search through huge amounts of data really quickly. It's used to find similar items by comparing their "vectors" instead of exact words. For example, it can help a chatbot find relevant documents even if the wording is slightly different.

5) Vector

A vector is just a way to represent information as numbers. In AI, text or images are often converted into vectors so that computers can compare and process them. It's like turning words into coordinates in a multi-dimensional space so that similar things are close together.

6) VectorDB

VectorDB is a type of database that stores these vectors. It's used to quickly find items that are similar in meaning or content. For example, if you ask a question, the AI can search the VectorDB to find the most relevant documents based on vector similarity.

7) Generative AI (GenAI)

Generative AI is AI that creates new content instead of just analyzing or classifying it. This could be text, images, music, or even videos. The cool part is that it can produce something completely new, like writing a poem or designing a new image from scratch.

8) GANs (Generative Adversarial Networks)

GANs are a special type of generative AI. They work by having two models compete with each other: one generates fake data, and the other tries to detect if it's fake. Over time, this competition makes the generator really good at creating realistic content, like photos that look like real people or objects.